built once per building

1–2 SENSE

the building's own data, turned into events

Trend data

1

What the building-automation system already records: temperatures, flows, damper and valve positions, fan speeds, once a minute. No new sensors.

Configuration file

1

Maps sensor names to roles and lists the control sequence as rules. The only building-specific part: a new building means a new file, not code.

Events 2

one case per day, rebuilt every new day

State events: what the equipment is doing

Signature events: what should not happen

state events only

signature events skip the models and go only to the Rules check

3–4a UNDERSTAND

what a normal day looks like, learned from one fault-free period

Process models

3

Discovered from the state events only, so they cannot learn the configured fault rules. One per terminal unit; one for the whole unit.

Calibration

4a

Healthy bands and model thresholds for every check come from the fault-free days. No threshold ever sees fault data.

Held-out days

4a

The last eight days of each month (96 a year) are kept aside and used only to measure each check's false-alarm rate.

run on every new day

4b CHECK

eight checks score every new day, each against its own threshold

Unit conformance	Terminal-unit conformance	Absence	Frequency	Rate	Oscillation	Residual	Rules
day fits the unit model	day fits the terminal-unit model	expected terminal unit silent	daily count out of range	last 30 days against same month; support only	actuator reversing too often	physical relationship drifted	a control rule broken: a signature event

read the event log

read the raw sensors

5–6 ALERT

only what beats its own false-alarm rate reaches the operator

Noise gate

5

A check may alarm only when its flags exceed what its own false-alarm rate would produce by chance. The rate is never taken below 3 days in 96.

Alarm to the operator

6

What broke. Where: unit, terminal unit, zone.
The evidence: values and duration.
The check's measured false-alarm rate.

STRATA

a process-mining framework for HVAC fault detection